

Lecture Slide - 2
Taylor Series &
Calculating Truncation Error

The Taylor Series

- The Taylor theorem states that any smooth function can be approximated as a polynomial.
- The Taylor series provides a means to express this idea mathematically.

$$f(x) = f(x_0) + \frac{x - x_0}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + R_n$$

Taylor Series

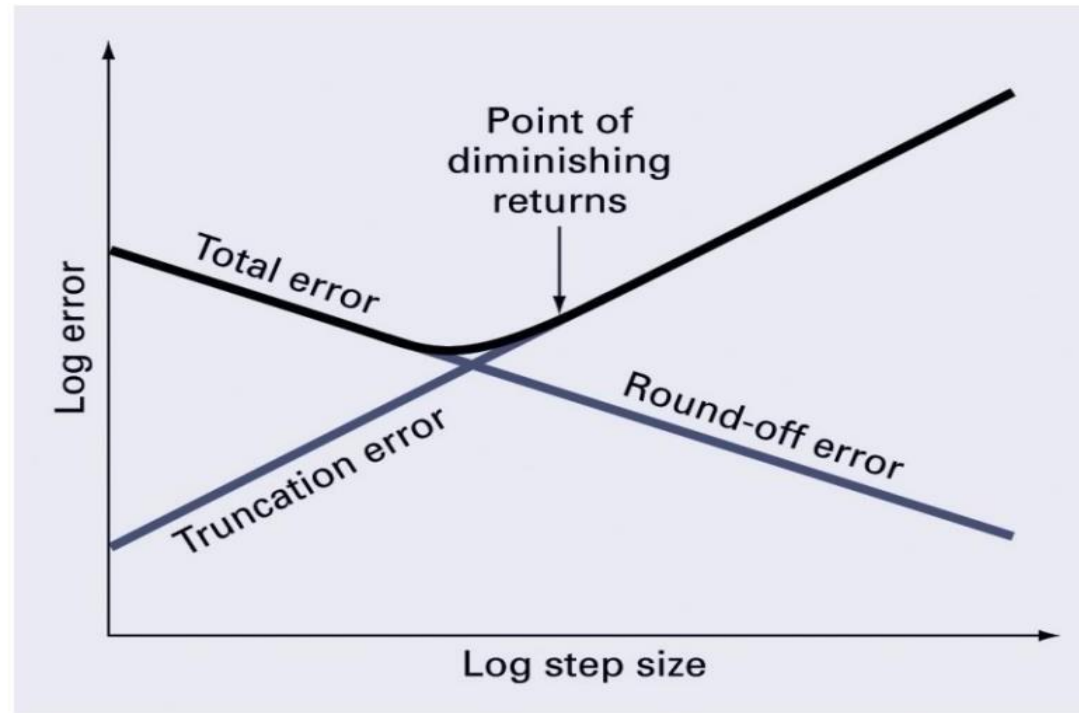
- Let $x=x_{i+1}$, $x_0=x_i$ and $(x_{i+1} - x_i) = h$, then the above series can be written as,

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

- In general, the n_{th} order Taylor series expansion will be exact for an n_{th} order polynomial.
- In other cases, the remainder term R_n is of the order of h^{n+1} , meaning:
 - The more terms are used, the smaller the error, and
 - The smaller the spacing, the smaller the error for a given number of terms

Total Numerical Error

- The **total numerical error** is the summation of the **truncation** and **roundoff errors**.
- The truncation error generally increases as the step size increases, while the roundoff error decreases as the step size increases - this leads to a point of diminishing returns for step size



Applications of Taylor expansion

□ For practice we denote $h = x - x_0$

□ Find the Taylor expansion of $\sin(x)$ about $x_0 = \pi/2$

Answer:

Derivative $d(\sin(x)) = \cos(x)$ and $d(\cos(x)) = -\sin(x)$

$$\begin{aligned}\sin(x) = & \sin(\pi/2) + h \cos(\pi/2) - \frac{h^2}{2!} \sin(\pi/2) - \frac{h^3}{3!} \cos(\pi/2) \\ & + \frac{h^4}{4!} \sin(\pi/2) + \frac{h^5}{5!} \cos(\pi/2) - \dots\end{aligned}$$

where $h = x - \pi/2$

Derivative $d(\tan(x)) = \sec^2(x)$ and $d(\sec^2(x)) = 2 \sec^2(x) \tan(x)$

Find the Taylor expansion of $\tan(x)$ about $x_0 = \pi/2$

Applications of Taylor expansion

- With the help of the previous page;
 - Find the Taylor expansion of $\cos(x)$ about $x_0 = \pi/3$
- Hints: $h = x - x_0$, $d(\sin(x)) = \cos(x)$ and $d(\cos(x)) = -\sin(x)$

Answer:

Taylor Series Approximation of a Polynomial

Use zero through fourth-order Taylor series expansions to approximate the function

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

from $x_i = 0$ with $h = 1$. That is, predict the function's value at $x_{i+1} = 1$.

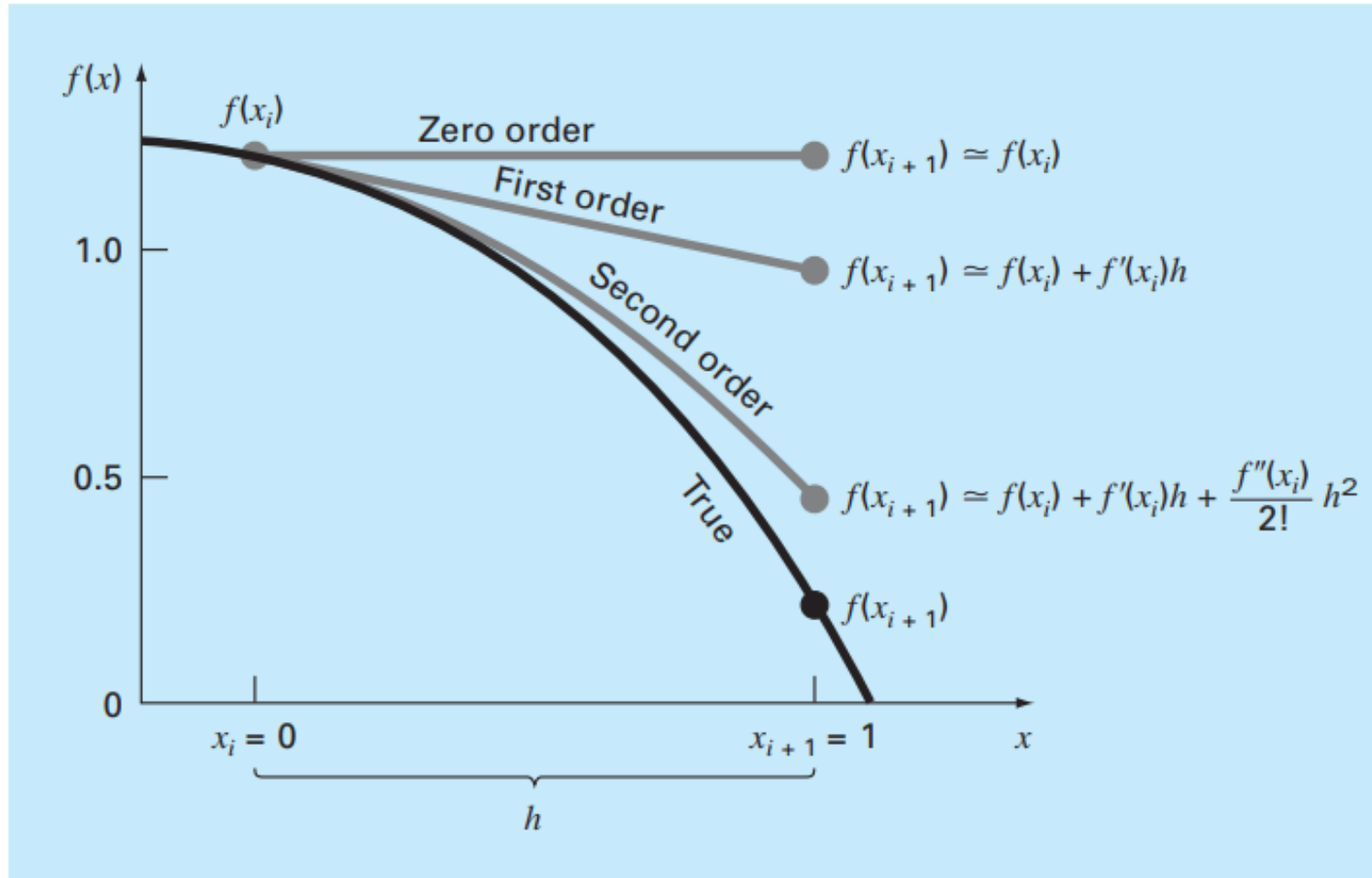
Because we are dealing with a known function, we can compute values for $f(x)$ between 0 and 1.

The function starts at $f(0) = 1.2$ and then curves downward to $f(1) = 0.2$. Thus, the true value that we are trying to predict is 0.2.

The Taylor series approximation with $n = 0$ is

$$f(x_{i+1}) \simeq 1.2$$

Taylor Series Approximation of a Polynomial



Taylor Series Approximation of a Polynomial

Transaction Error is

$$E_t = 0.2 - 1.2 = -1.0$$

at $x = 1$.

For $n = 1$, the first derivative must be determined and evaluated at $x = 0$:

$$f'(0) = -0.4(0.0)^3 - 0.45(0.0)^2 - 1.0(0.0) - 0.25 = -0.25$$

Therefore, the first-order approximation is $f(x_{i+1}) \approx 1.2 - 0.25h$

which can be used to compute $f(1) = 0.95$.

Consequently, the approximation begins to capture the downward trajectory of the function in the form of a sloping straight line.

This results in a reduction of the truncation error to $E_t = 0.2 - 0.95 = -0.75$

Taylor Series Approximation of a Polynomial

For $n = 2$, the second derivative is evaluated at $x = 0$:

$$f''(0) = -1.2(0.0)^2 - 0.9(0.0) - 1.0 = -1.0$$

Therefore, $f(x_{i+1}) \approx 1.2 - 0.25h - 0.5h^2$

and substituting $h = 1$, $f(1) = 0.45$.

The inclusion of the second derivative now adds some downward curvature, resulting in an improved estimate.

The truncation error is reduced further to $0.2 - 0.45 = -0.25$.

Taylor Series Approximation of a Polynomial

Taylor Series can be written as,

$$v(t_{i+1}) = v(t_i) + v'(t_i)(t_{i+1} - t_i) + \frac{v''(t_i)}{2!}(t_{i+1} - t_i)^2 + \cdots + R_n$$

Now let us truncate the series after the first derivative term:

$$v(t_{i+1}) = v(t_i) + v'(t_i)(t_{i+1} - t_i) + R_1$$

Equation (4.13) can be solved for

$$v'(t_i) = \underbrace{\frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}}_{\text{First-order approximation}} - \underbrace{\frac{R_1}{t_{i+1} - t_i}}_{\substack{\text{Truncation} \\ \text{error}}} \\ \qquad \qquad \qquad \downarrow \\ \frac{R_1}{t_{i+1} - t_i} = O(t_{i+1} - t_i)$$

Thus, there is a truncation error of order $t_{i+1} - t_i$.

In other words, the error of our derivative approximation should be proportional to the step size.

Consequently, if we halve the step size, we would expect to halve the error of the derivative.

Numerical Differentiation

□ The first order Taylor series can be used to calculate approximations to derivatives:

◆ Given: $f(x_{i+1}) = f(x_i) + f'(x_i)h + O(h^2)$

◆ Then: $f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$

□ This is termed a “forward” difference because it utilizes data at i and $i+1$ to estimate the derivative.

Numerical Differentiation

□ There are also backward difference and centered difference approximations, depending on the points used:

□ Forward:

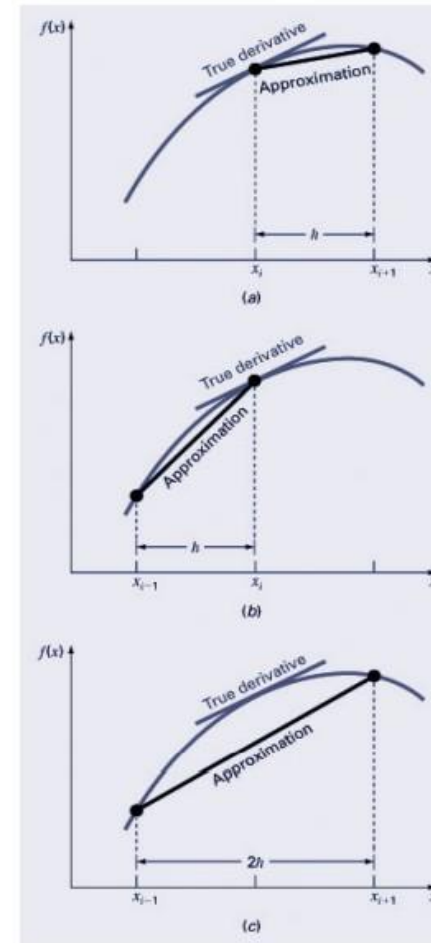
$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$$

□ Backward:

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$

□ Centered:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} + O(h^2)$$



Proof of *centered difference*
representation of the
first derivative from the
Book Page no 95

Calculating Truncation Error

Example 1:

Find the Taylor expansion of e^x about $x_0 = 0$ which use the first two, three, four and five and evaluate for $x = 0.5$, respectively.

Answer: Given, $x_0 = 0$, $\therefore h = (x - x_0) = x$

$$e^x = 1 + x + O(x^2)$$

$$e^x = 1 + x + \frac{x^2}{2!} + O(x^3)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + O(x^4)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + O(x^5)$$

Calculating Truncation Error

Example 1:

Find the Taylor expansion of e^x about $x_0 = 0$ which use the first two, three, four and five and evaluate for $x = 0.5$, respectively.

Answer: Given, $x_0 = 0$, $\therefore h = (x - x_0) = x$

$$e^x = 1 + x + O(x^2)$$

$$e^x = 1 + x + \frac{x^2}{2!} + O(x^3)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + O(x^4)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + O(x^5)$$

Calculating Truncation Error

$$e^{0.5} = 1.648721.. \text{ (True value)}$$

$$e^{0.5} = 1.5 + O(x^2) \quad \longrightarrow \quad O(x^2) \cong 0.14872..$$

$$e^{0.5} = 1.625 + O(x^3) \quad \longrightarrow \quad O(x^3) \cong 0.023721..$$

$$e^{0.5} = 1.64583.. + O(x^4) \quad \longrightarrow \quad O(x^4) \cong 0.002887..$$

$$e^{0.5} = 1.64843.. + O(x^5) \quad \longrightarrow \quad O(x^5) \cong 0.0002837..$$

Calculating Truncation Error

Example 2:

Find the Taylor expansion of e^x about $x_0 = 0.25$ which use the first two, three, four and five and evaluate for $x = 0.5$, respectively.

Answer: Given, $x_0 = 0.25$, $\therefore h = (x - x_0) = (0.5 - 0.25) = 0.25$

$$e^x = e^{0.25} + he^{0.25} + O(h^2) \quad \text{since } (d(e^x))/(dx) = e^x$$

$$e^x = e^{0.25} + he^{0.25} + \frac{h^2}{2!} e^{0.25} + O(h^3)$$

$$e^x = e^{0.25} + he^{0.25} + \frac{h^2}{2!} e^{0.25} + \frac{h^3}{3!} e^{0.25} + O(h^4)$$

$$e^x = e^{0.25} + he^{0.25} + \frac{h^2}{2!} e^{0.25} + \frac{h^3}{3!} e^{0.25} + \frac{h^4}{4!} e^{0.25} + O(h^5)$$

Calculating Truncation Error

$$e^{0.5} = 1.648721..$$

$$e^{0.5} = 1.60503.. + O(h^2) \longrightarrow O(h^2) \cong 0.04368..$$

$$e^{0.5} = 1.64515.. + O(h^3) \longrightarrow O(h^3) \cong 0.00356..$$

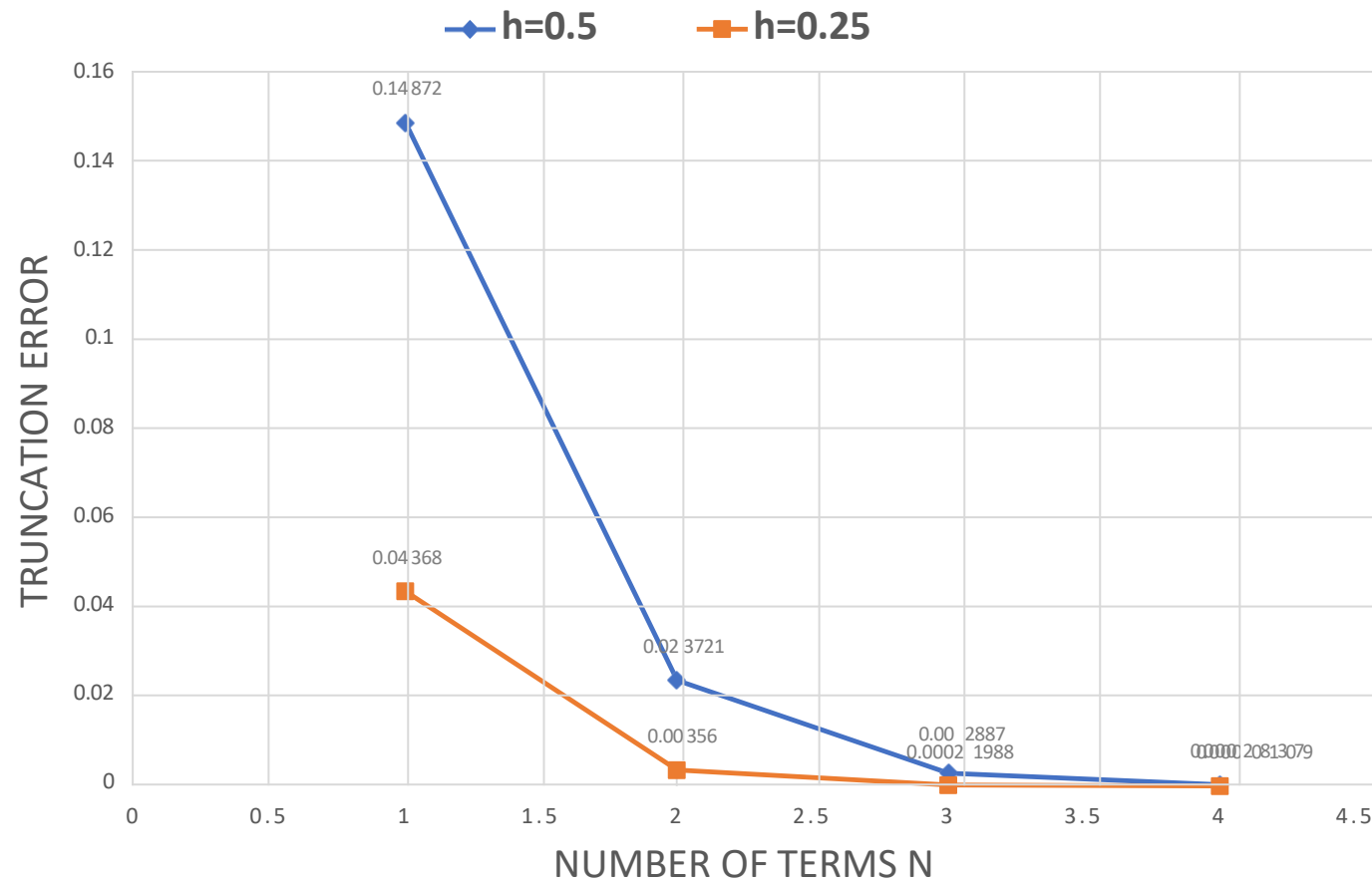
$$e^{0.5} = 1.64850.. + O(h^4) \longrightarrow O(h^4) \cong 2.1988..e-004$$

$$e^{0.5} = 1.64871.. + O(h^5) \longrightarrow O(h^5) \cong 1.0900..e-005$$

Summary from example 1 and 2:

<i>Order of Trunc. Er.</i>	$x_0=0, h=0.5$	$x_0=.25, h=0.25$	<i>Ratio</i>
$O(h^2)$	0.14872..	0.04368..	3.4
$O(h^3)$	0.023721..	0.00356..	6.66
$O(h^4)$	0.002887..	$2.1988.. \times 10^{-4}$	13.13
$O(h^5)$	0.0002837..	$1.090.. \times 10^{-5}$	26.03

Truncation error vs n



More Example

Use forward and backward difference approximations of $O(h)$ and a centered difference approximation of $O(h^2)$ to estimate the first derivative of

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

at $x = 0.5$ using a step size $h = 0.5$. Repeat the computation using $h = 0.25$.

the derivative can be calculated directly as

$$f'(x) = -0.4x^3 - 0.45x^2 - 1.0x - 0.25$$

and can be used to compute the true value as $f'(0.5) = -0.9125$.

More Example

Solution. For $h = 0.5$, the function can be employed to determine

$$\begin{aligned}x_{i-1} &= 0 & f(x_{i-1}) &= 1.2 \\x_i &= 0.5 & f(x_i) &= 0.925 \\x_{i+1} &= 1.0 & f(x_{i+1}) &= 0.2\end{aligned}$$

These values can be used to compute the forward divided difference

$$f'(0.5) \cong \frac{0.2 - 0.925}{0.5} = -1.45 \quad |\varepsilon_t| = 58.9\%$$

The backward divided difference is

$$f'(0.5) \cong \frac{0.925 - 1.2}{0.5} = -0.55 \quad |\varepsilon_t| = 39.7\%$$

The centered divided difference is

$$f'(0.5) \cong \frac{0.2 - 1.2}{1.0} = -1.0 \quad |\varepsilon_t| = 9.6\%$$

For $h = 0.25$,

$$\begin{aligned}x_{i-1} &= 0.25 & f(x_{i-1}) &= 1.10351563 \\x_i &= 0.5 & f(x_i) &= 0.925 \\x_{i+1} &= 0.75 & f(x_{i+1}) &= 0.63632813\end{aligned}$$

which can be used to compute the forward divided difference,

$$f'(0.5) \cong \frac{0.63632813 - 0.925}{0.25} = -1.155 \quad |\varepsilon_t| = 26.5\%$$

the backward divided difference,

$$f'(0.5) \cong \frac{0.925 - 1.10351563}{0.25} = -0.714 \quad |\varepsilon_t| = 21.7\%$$

and the centered divided difference,

$$f'(0.5) \cong \frac{0.63632813 - 1.10351563}{0.5} = -0.934 \quad |\varepsilon_t| = 2.4\%$$

The centered difference approximation is more accurate than forward or backward differences.

More Example

Use Taylor series expansions with $n = 0$ to 6 to approximate $f(x) = \cos(x)$ at $x_{i+1} = \pi/3$ based on the value of $f(x)$ and its derivatives at $x_i = \pi/4$. Note that this means that $h = \pi/3 - \pi/4 = \pi/12$.

Ans.: $f(\pi/3) = \cos(\pi/3) = 0.5$ (True value)

➤ Considering result on 1st value only:

➤ $f(\pi/3) \cong \cos(\pi/4) = 0.707106781$

➤ which represents a percent relative error of

$$\varepsilon_t = ((|0.5 - 0.707106781|)/0.5) \times 100\% = 41.4213562\%$$

➤ Taking 1st 2 values:

➤ $f(\pi/3) \cong \cos(\pi/4) - \pi/12 \times \sin(\pi/4)$

$$= 0.707106781 - 0.2617993878 \times 0.707106781 = 0.521986658625$$

$$\varepsilon_t = ((|0.5 - 0.521986658625|)/0.5) \times 100\% = 4.397331725\%$$

Example ...

➤ Taking 1st 3 values:

$$\begin{aligned} f(\pi/3) &\cong \cos(\pi/4) - \pi/12 \times \sin(\pi/4) - (\pi/12)^2/2 \times \cos(\pi/4) \\ &= 0.707106781 - 0.2617993878 \times 0.707106781 \\ &\quad - (0.2617993878)^2 \times 0.707106781/2 = 0.4977544913 \end{aligned}$$

$$\mathcal{E}_t = ((|0.5 - 0.4977544913|)/0.5) \times 100\% = 0.4491017457\%$$

Example ...

n	$f(\pi/3)=0.5(True)$	% Error
0	0.707106781	41.4213562
1	0.521986659	4.397331725
2	0.497754491	0.4491017457
3	0.499869147	2.62×10^{-2}
4	0.500007551	1.51×10^{-3}
5	0.500000304	6.08×10^{-5}
6	0.499999988	2.44×10^{-6}

Example ...

